PROJECT GOVERNANCE, RISK & CLOSURE PLAN

MBA 666

Green Hydrogen Production Plant

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Document Purpose: To define the control framework, risk response plan, and formal procedures for closing the project, supporting the optimal 36-week timeline.

I Project Governance

A. Overall Project Structure (Command Chain)

We use a Hierarchial Matrix approach which is common in Time constrained projects to make clear reporting and ensuring team autonomy is still there.

Position	Level of Authority	Primary Focus
Project Sponsor	Executive	Strategic oversight, budget approval, major risk owner.
Project Manager (PM)	Functional	CPM adherence, resource allocation (levelling), risk response.

Team Leads	Operational	Task execution, quality control, daily resource use, inter-team handoffs.
Field/Specialist Personnel	Execution	Daily physical/technical work.

B. Individual Team Structures (Resource Pools)

The project relies on three primary resource pools, each structured to execute specific activities efficiently. The resource demand for the Field Teams (HR-C and HR-I) is restricted to 1 unit per team.

1. Civil Team (HR-C) – 1 Unit

Role: Executes foundation and site preparation work (Activity B).
Composition:

Component	Quantity	Primary Function
Team Lead	1	Manages daily schedule and safety.
Civil Foreman	1	Direct supervision of site foundation work.
Skilled Labor	3	Execution of Activity B.

Team Lead Responsibility: Managing the schedule for Activity B and confirming readiness for the seamless handoff to the HR-I team at Week 12 (Levelled Schedule).

2. Install Team (HR-I) – 1 Unit

Role: Executes mechanical and electrical installation/piping (Activities E and F)
Composition:

Component	Quantity	Primary Function
Team Lead	1	Manages mechanical and electrical installation tasks.
Tech. Supervisors	1	Oversees specialized equipment placement.
Skilled Labor	3	Execution of Activities E, F and G.

Team Lead Responsibility: Receiving the site from HR-C at Week 12 and ensuring continuous operation through Week 26.

3. Specialist Team (Task Specific)

Role: Executes critical, unique, or complex activities (eg. procurement, H_2 system integration, final testing).
Composition:

Specialist Role	Assigned Activity	Duration (Wk)	Key Task
Procurement	A (4), C (12)	16	Ensure all H2 components and long-lead items are secured by **EF = 16** (Activity C LF), as C has zero slack.
Engineering/Design	H(12)	12	Final design review and integration of the hydrogen system (Zero Slack).
SCADA/Automation	I(4),J(2)	6	Execute system testing, validation, and commissioning on schedule (Zero Slack).
Project Control	K(2)	2	Final documentation, handover, and formal project closure

Specialist Lead Responsibility: Ensuring all critical path activities (A, C, F, H, I, J, K) start and finish on their ES and EF dates due to their zero slack.

C. Escalation Matrix

Severity Level	Trigger Condition	Action/Mitigation Strategy	Authority
Level 1 (Operational)	Local schedule variance within activity slack.	Utilize slack; resolve conflict locally.	Team Lead -> PM

Level 2 (Schedule/Resource Threat)	Loss of a key HR or delay consuming 50 % of an activity's slack.	Initiate Resource Constraint Contingency Plan (Section 5 in the master document).	Project Manager (PM)
Level 3 (Major Breach)	Confirmed deviation from the Critical Path (Zero Slack) or potential budget overrun.	Authorize project re-baselining or stop work order.	Project Sponsor

II Risk Assessment and Mitigation Strategy

A. Overall Project Structure (Command Chain)

The primary operational risk has already been mitigated through the resource levelling calculated in Resource Planning Document

Risk Factor	Impact	Mitigation Strategy
Resource Demand Fluctuation	Increased indirect costs due to resource idle time or double mobilization (the Week 11gap).	Utilize 1 week of slack from Activity B to shift the **HR-C** schedule. This ensures a seamless handoff to **HR-I** at Week 12, eliminating the idle period and achieving continuous site activity from Week 6–25 and week 17-26.
Handoff Failure	Delay to the start of Activity G and subsequent **HR-I** work.	**PM** and **HR-C and HR-I Team Leads** hold mandatory sign-off meetings at Week 11 to confirm Activity B completion and **G** readiness.

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B.Contingency Plan: Resource Constraint Heuristics

Since the project is currently time-contrained any risk event could force the projects into a resource constrained environment.

If this happens ,project team cannot level and should implement the parallel method using specific heuristic priority rules to manage resource conflicts.This serves as the primary **mitigation strategy** for resource loss.

Priority Rank	Heuristic Rule	Purpose
1	Minimum Slack	Prioritizes the activity with the least total float. This is the **most time-critical** task and is chosen first to protect the 36-week deadline
2	Smallest Duration	Prioritizes the task that can be completed fastest. This strategy aims to **clear the limited resource bottleneck** in the shortest possible time
3	Lowest Activity ID number	Used as a final, objective tie-breaker when multiple activities have equal slack and duration.

C.Critical Path Risk Management

Since the critical path activities (A,C,F,H,I,J,K) have zero slack, any delay is an immediate threat to the project deadline. Mitigation relies on rigorous monitoring and the Specialist Team's accountability.

Risk Factor	Impact	Mitigation Strategy
Procurement Delay (Activity C)	1-to-1 delay of the entire project timeline.	• Specialist Lead monitors supplier LF dates against the EF of Week 16. • Proactive communication to PM if risk exceeds 5 days.
Integration Failure (Activity H)	Rework/delay in final H_2 system integration.	• Dedicated Engineering Specialist assigned to H to ensure quality control. • Escalation is Level 2 (PM) immediately upon H starting without final design sign-off.
Testing Failure (Activities I, J)	Failure to meet functional requirements; project closure delay.	SCADA/Automation Specialist required to provide weekly progress reports against the Zero Slack schedule

III Risk Assessment and Mitigation Strategy

For this project closure is reached upon achieving both technical + physical + administrative + contractual conditions.
The overall project completion is tied to the successful execution of the final zero-slack activities on the Critical Path (I , J, K).

A. Technical and Physical Closure conditions

Closure Condition	Requirement	Responsibility	Final CPM Activity
System Validation	Final testing of all integrated H_2 generation systems, control loops, and safety protocols (SCADA and Automation).	SCADA Specialist Lead	Activity I (Wk 42)
Functional Acceptance	Final plant commissioning demonstrating that the facility meets all specified 1 MW generation and integration requirements.	Project Manager (PM)	Activity J (Wk 44)
Physical Handover	Completion of site clearance, demobilization of all field teams (HR-C / HR-I), and formal transfer of the physical asset to the owner or operations team.	Field Team Leads	Activity K (Wk 46)
Documentation Handover	Compilation and submission of all As-Built drawings, maintenance manuals, and operational protocols to the owner.	Specialist Lead	Activity K (Wk 46)

B. Administrative and Contractual Closure

These conditions ensure all financial, legal, and historical project obligations are met.

Financial Reconciliation

- All vendor invoices,including the final payment for long lead activities(Activity C) are processed and reconciled against the final project budget
- The final budget status is reported to the Project Sponsor.
- Contingency reserves are closed out.

Contractual Release

- Final completion certificates are issued to all major contractors.

- Contractual obligations for the Field Teams (HR-C/HR-I) are formally terminated.
- Resources are officially released from the project.
- Warranty periods for installed equipment are formally activated.

Knowledge Management

- A *Lessons Learned* session is conducted with all Team Leads and the PM.
- The final *Project Resourcing, Governance, Risk & Closure Plan* document, along with the final CPM and resource charts, is archived in the central project repository.